

Mechanical loading and adaptive responses of tendinous tissues

Falk Mersmann, Sebastian Bohm and Adamantios Arampatzis

Mechanisms of tendon plasticity

Tendons transmit the forces generated by the muscles to the skeleton to produce movement. The elasticity of tendons, which allows tendons to deform (mostly elongate) upon load application, yet restore their original length when the load is released, importantly contributes to movement performance, since it influences the muscle fascicle behavior. As the muscle adapts well to exercise, it seems plausible that tendons feature a certain degree of plasticity as well. It has been shown already in the 1940s, by comparing tendons from rats with a different degree of habitual loading, that tendinous tissue is sensitive to mechanical stimulation (Ingelmark, 1945). However, it took some major technological and methodological developments to address the mechanisms of human tendon adaptation *in vivo*. Research from the past two decades has provided convincing evidence that the tendinous tissue of humans adapts over a whole lifespan. When a tendon is repeatedly subjected to increased mechanical loading, for instance in the course of resistance exercise, the associated increases of muscle strength are commonly accompanied by an increase of tendon stiffness (Kubo et al., 2001; Arampatzis et al., 2007). Tendon stiffness is defined as the relationship of applied force and resultant elongation and, thus, describes the capability of a tendon to resist force (Figure 5.1).

Two main mechanisms may account for the increase of stiffness in response to loading. It seems that, especially in the early stages of tendon adaptation, structural changes of the extracellular matrix influence the material properties of the tissue and increase the internal mechanical integrity of the collagenous network. These potential changes could include the type and degree of intra- and interfibrillar collagen cross-linking (Depalle et al., 2015), glycosaminoglycan content contributing to interfibrillar connections and force transfer (Cribb and Scott, 1995), and collagen area fraction (Robinson et al., 2004). Although the tendon core tissue is supposed to be relatively inert (Heinemeier et al., 2013), the elastic modulus of tendons—as a measure of their material properties based on the stress–strain relationship—might increase by up to 85% following 10–14 weeks of training (Wiesinger et al., 2015).

Hypertrophy of the tendon in response to loading, on the other hand, has been observed less frequently and is now rather considered a long-term mechanism contributing to increased tendon stiffness (Bohm et al., 2015). The radial growth probably occurs via the synthesis of new extracellular matrix in the outer layers of the tendon (Gumucio et al., 2014). It seems unlikely that a significant increase of the tendon cross-sectional area occurs much earlier than after 3 months of training, and

Chapter 5

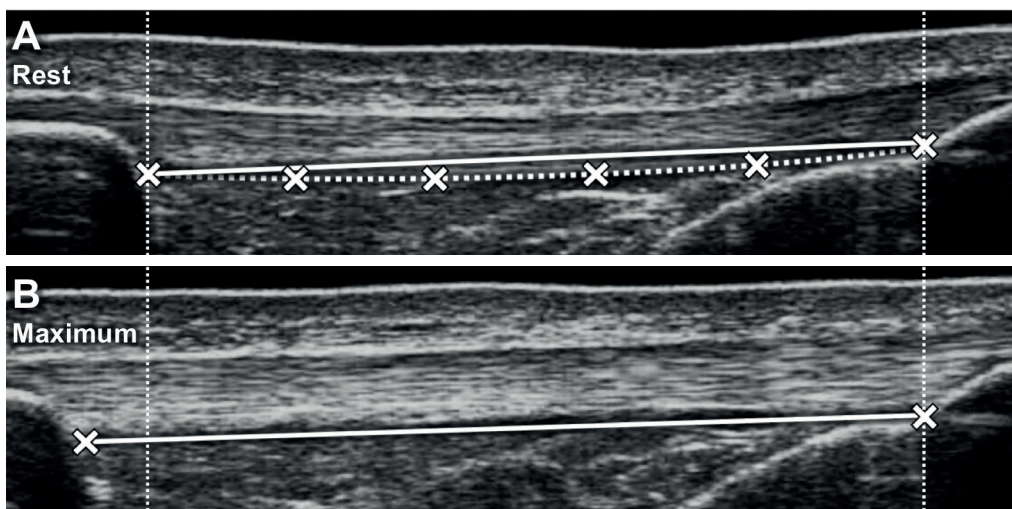
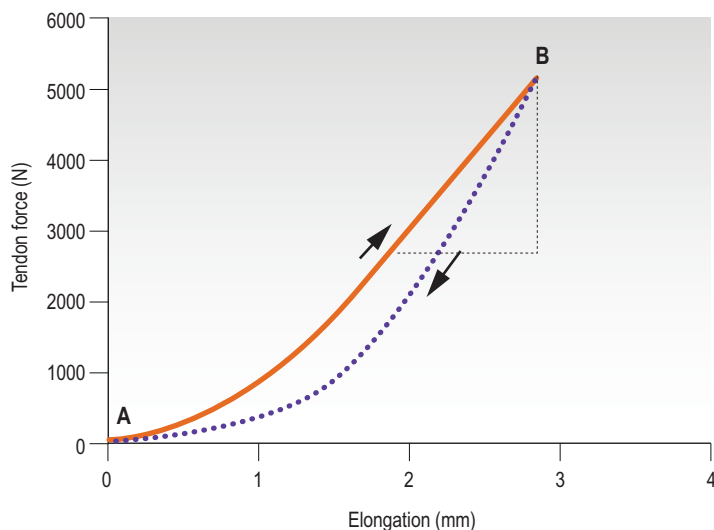


Figure 5.1 Exemplary force-elongation behavior and ultrasound images of a patellar tendon during loading from rest

(A) to a maximum voluntary muscle contraction (B). The force-elongation relationship is characterized by a flat so-called "toe region", where only low forces are needed to stretch the tendon due to the wavy arrangement of the collagen fibrils and fibers, and a subsequent linear increase. The slope of this increase (indicated by the pointed triangle) is referred to as "stiffness" and describes the resistance of a tendon against elongation.

its development appears to be site-specific (i.e. in some regions of the tendon, its cross-section increases while remaining unaffected in others; Arampatzis et al., 2007; Kongsgaard et al., 2007). The radial tendon growth reported

in longitudinal studies on human Achilles or patellar tendons is usually moderate in magnitude; however, years of training might lead to more marked increases of the order of up to 35% (Wiesinger et al., 2015).

Mechanical loading and adaptive responses of tendinous tissues

It is still poorly understood how the metabolic responses of tendons, structural, morphological and mechanical changes of tendinous tissue associate with one another. The temporal development of the adaptation process suggested above clearly needs further confirmation regarding the methodological challenges and simplified assumptions associated with *in vivo* investigations. Nevertheless, it is evident that tendons adapt to systematic loading with an increase of stiffness and it is likely that changes of the material properties precede tendon hypertrophy.

Relevance of tendon training

As described above, the long-term adaptation of tendon stiffness alongside the development of muscular strength may serve as a protective mechanism against injury. The ultimate strain of tendons is considered to be relatively constant (LaCroix et al., 2013), and a sole increase of muscle strength would progressively pull the tendon to levels of strain closer to and finally above its physiological threshold upon maximum effort muscle contractions. However, within a training process it might be that the balance of muscle strength and tendon stiffness gets distorted. The tissue renewal rate of tendons (turnover) is lower compared with muscle tissue (Heinemeier et al., 2013), which can result in a delayed adaptation of the tendon in early stages of training (Kubo et al., 2010; 2012) and may explain the high prevalence of overuse injuries (e.g. tendinopathy) when training loads are changed rapidly (Gabbett, 2016). Moreover, it seems that not all types of loading that induce an increase of muscle strength are equally effective for the tendon. For instance, it is well accepted that training with moderate loads until fatigue can increase muscle strength and size (Ozaki et al., 2016), while no such reports

exist for tendon stiffness. In fact, an increase of muscle strength and no change of stiffness have been observed after isometric training at 55% of the voluntary maximum (Arampatzis et al., 2007; 2010). Such asynchronous adaptation results in an increase of tendon strain and thus a higher mechanical demand for the tendon (as illustrated in the left panel of Figure 5.2), which is considered a risk factor for tendon overuse injury (Wren et al., 2003). Further, it has been demonstrated in adolescent volleyball athletes that sport-specific loading (i.e. jumping) causes a greater increase of muscle strength compared with tendon stiffness. This, in turn, leads to chronically increased levels of tendon strain during maximum effort muscle contractions (Mersmann et al., 2017b) and episodes of exceptionally high values over the course of a competitive season (Mersmann et al., 2016). Considering the dramatic prevalence of tendinopathy in athletes from jump disciplines (Lian et al., 2005), it becomes evident that it could be an important aspect of injury prevention to provide specific training stimuli to strengthen the tendon for a balanced musculotendinous development, which maintains tendon strain at physiological levels despite increased muscle strength capacity (Figure 5.2, right panel).

In addition to the prevention of overuse injury, tendon training also plays a key role in the rehabilitation of tendinopathy. Classic loading programs, such as Alfredson's eccentric training (Chapter 16), have been shown to be effective in reducing pain and improving function (Malliaras et al., 2013), yet the moderate intensity nature of loading appears insufficient to address the mechanical weakening of the tendon (Arya and Kulig, 2010) by increasing its stiffness (Foure et al., 2013). More recent approaches which use loading paradigms characterized by a high

Chapter 5

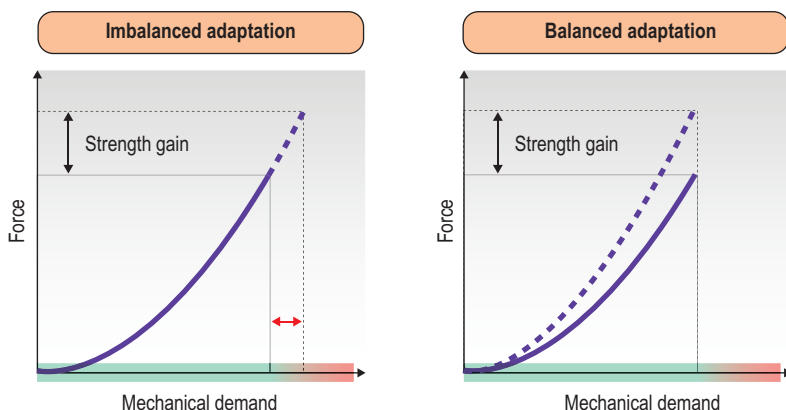


Figure 5.2

If the force that can be exerted by a muscle increases without a similar increase of tendon stiffness (left), the mechanical demand for the tendon increases to potentially detrimental levels (red zone), which could pose a risk of overuse injury. On the other hand, by adequately stimulating tendon adaptation (right), the mechanical demand for the tendon remains in physiological bounds (green), despite the greater absolute load it is subjected to.

loading intensity (e.g. heavy-slow resistance training) seem more suitable in this regard (Couppé et al., 2015).

The mechanical properties of tendons also greatly influence performance in everyday activities and sports. Since tendons can store and release elastic strain energy, they affect the operating behavior and force potential of muscles and the power output of the muscle-tendon unit (Roberts, 2016). In humans, muscle fascicle operating behavior seems optimized for metabolic economy, for example in running (Bohm et al., 2018), and for power production in jumping (Nikolaidou et al., 2017). A sole increase of muscle strength due to training could interfere with this interaction of muscle and tendon, limiting the extent to which the increased muscle capacity translates into performance. Moreover, high tendon stiffness positively affects the rate of force development and thus jump height or balance recovery, for example, after sudden perturbations (Bojsen-Møller et al., 2005; Karamanidis et al., 2008). Therefore, knowledge of effective loading paradigms to change the mechanical properties of tendons is also essential for the improvement of physical performance.

How to stimulate tendon adaptation

As described in Chapter 4, the adaptation of tendons is based on the transmission of the external tissue strain to the cytoskeleton of the embedded fibroblasts. The conformational changes of transmembrane proteins that occur during load application and the activation of stretch-sensitive ion channels in the cell membrane then trigger intracellular signaling cascades, which stimulate gene and growth factor expression for the upregulation of collagen and matrix protein synthesis (Wang, 2006).

From a mechanobiological point of view, there are four main factors which determine the transduction of the mechanical stimulus applied to the tendon into cellular signals: strain magnitude, strain rate, strain duration and strain frequency. First, the magnitude of strain applied to the tendon during loading directly determines the deformation of the fibroblasts (Arnoczky et al., 2002), which in turn seems to regulate the metabolic activity in response to loading (Lavagnino et al., 2003). It has been further suggested that the recruitment of collagen fibers is also greater during high strain compared with low strain loading (Kastelic et al., 1980), which

Mechanical loading and adaptive responses of tendinous tissues

might increase the number of effectively stimulated fibroblasts. Human *in vivo* studies on the Achilles and patellar tendon confirmed the importance of high magnitude tendon strain for the adaptive response to exercise. Arampatzis et al. (2007; 2010) compared training protocols of high intensity (i.e. 90% isometric maximum voluntary contraction [MVC] corresponding to 4.6% of tendon strain) and moderate intensity loading (i.e. 55% MVC corresponding to 2.9% of tendon strain) with equal overall training volume. Significant changes of the Achilles tendon morphological and mechanical properties were observed only following the loading regimen that induced high tendon strain during muscle contraction. Of all the factors that might determine tendon adaptation, it is now most clearly established that high loading is necessary to achieve changes of the mechanical properties of tendinous tissues (Bohm et al., 2015). A common misunderstanding is, however, that eccentric loading is most effective to stimulate a tendon. Yet the strain the tendon is subjected to during a contraction does not depend on the type of muscle contraction (i.e. isometric, concentric, eccentric) but only on the force the muscle is exerting. Accordingly, scientific evidence shows that the contraction type does not per se influence the loading response of tendons (Kjaer and Heinemeier, 2014). Nevertheless, due to the high force potential and economic force production of muscles during eccentric contractions, the exercise mode is a feasible way to provide repetitive high tendon strain with comparatively low muscle fatigue. The necessity of high tendon forces for initiating adaptive responses in the tendon also explains why stretching, against common belief, does not change the mechanical properties of tendons in the long term. As an inactive muscle has a lower stiffness compared with its tendon, the muscle belly takes over most of the length change during a passive stretch of the

muscle-tendon unit and the forces and strain in the tendon are quite low (De Monte et al., 2006). The increase in flexibility, which is commonly observed following acute or chronic stretching, seems more related to neural than mechanical changes (Freitas et al., 2018).

The strain rate (i.e. change in strain per unit of time) describes how fast strain is applied to a tendon. It has been shown that a high strain rate induces high fluid flow-related shear stress to the tenocytes, which is believed to trigger anabolic responses (Lavagnino et al., 2008). Further, the recruitment of collagen fibrils seems to increase with strain rate as well (Clemmer et al., 2010). In *in vivo* studies, high strain rates can be applied using plyometric loading through jumping exercises with short contact times. Interestingly though, the majority of these studies found no significant training effects on the mechanical properties of tendons or rather low effects. One possible explanation for these findings is the comparatively short duration of strain application with each loading cycle during plyometric loading, which seems to be another important factor for the mechanotransduction. In a series of systematic intervention studies, in which the overall loading volume (i.e. the integral of the force-time curve of the training sessions) was kept constant, high strain was applied to the Achilles tendon for <130 ms, 1 s, 3 s and 12 s per loading cycle, respectively (Arampatzis et al., 2010; Bohm et al., 2014). The strongest effects on tendon stiffness were observed following the 3 s loading protocol, which supports the idea that the duration of strain application is an important factor for tendon adaptation and that the mechanical stimulation might be too short to be effective in plyometric exercises or high frequency-short duration loading protocols. It seems possible that an increase of loading duration could lead to a more homogenous strain

Chapter 5

distribution and, thus, tenocyte stimulation within the tendon. A longer duration of tenocyte deformation might also show effects on the cellular signaling pathways, for example, the influx and propagation of calcium within and between the tenocytes (Lavagnino et al., 2015). These hypotheses (i.e. more homogeneous tenocyte stimulation and more effective signaling) do not contradict that fluid flow-induced shear stress at the cytoskeleton of the tenocytes is an important stimulus yet, under *in vivo* loading conditions, the benefit of a high compared with a low strain rate might be offset by a compromised mechanotransduction due to a reduced strain duration in certain modes of loading. However, our current understanding of the effects of specific types of mechanical stimulation of the tendon on the complex cellular machinery and signaling pathways of mechanotransduction is widely deficient, and the main conclusion that can be drawn from tendon adaptation research is that an effective stimulus for the tendon needs to exceed a certain threshold in terms of strain magnitude and, supposedly, strain duration.

Other loading determinants for tendon training are, to date, widely unclear. The effect of strain frequency, which is determined by both load and rest duration in cyclic loading, has been investigated in humans *in vivo* first by Arampatzis et al. (2010). The results suggested a lower strain frequency (0.17 Hz) to be more effective compared with a higher one (0.5 Hz). However, differences in the strain duration (3 s and 1 s in the low and high frequency protocol, respectively) might also partially account for these results. Waugh et al. (2018) compared two high intensity isometric training protocols, differing only in the rest duration between loading cycles of either 3 s (i.e. 0.17 Hz) or 10 s (i.e. 0.08 Hz), respectively. While the effect on the mechanical properties of the Achilles tendon were of similar magnitude,

some structural changes that might be interpreted as early indications of overload were detected only following the protocol with the short rest duration. However, it needs to be acknowledged that these findings might be exclusive to a training regimen of such an overall loading volume, which was markedly higher (by factor 1.9) compared with earlier work using high loading intensities (Arampatzis et al., 2010; Bohm et al., 2014). Nevertheless, it might be concluded that the results on the effects of strain frequency are in principal in accordance with reports from *in vitro* models, suggesting that intermediate levels of strain frequency between 0.1 and 0.25 Hz provide an effective stimulus (Joshi and Webb, 2008). The optimal duration of rest between training sessions, however, is even more unclear. Similarly, the effect of weekly training volume has not been investigated thus far.

Clinical summary

Although several factors influencing the effectiveness of tendon loading protocols are not well investigated, some recommendations can be given on how to increase the stiffness of a tendon (see also Figure 5.3). A tendon training program should be characterized by an intensity of $\geq 85\%$ of the isometric or one-repetition maximum and the duration of the muscle contraction around 3 s, while the contraction mode (i.e. isometric, concentric, eccentric) does not play a major role. Isometric training should be performed in joint angles close to the optimum for force generation (i.e. $\sim 60^\circ$ knee flexion for patellar tendon training or $\sim 10^\circ$ ankle dorsiflexion and extended knee for the Achilles tendon). This has the advantage of being easy to learn and the load can be controlled precisely. During dynamic exercises (i.e., eccentric-concentric movements), high tendon forces occur only in specific ranges of joint angles during movement (e.g. between 60°

Mechanical loading and adaptive responses of tendinous tissues

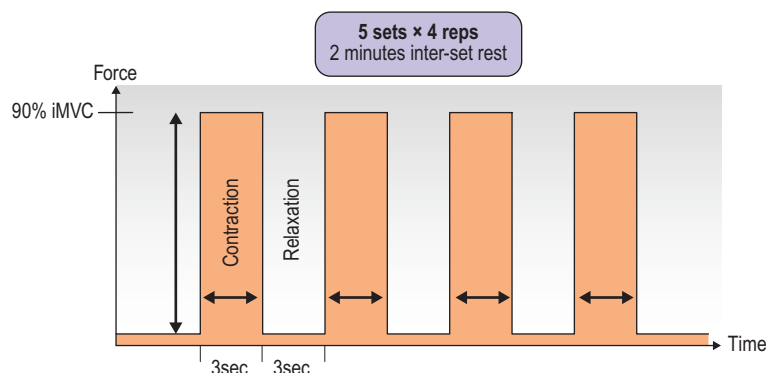


Figure 5.3

Evidence-based recommendations for a potential structure of tendon training. High intensity loading to the tendon (85–90% iMVC) should be applied in five sets of four repetitions with a contraction and relaxation duration of 3 s, and a rest between sets of 2 min. The training is to be applied 3–4 times a week for at least 12 weeks to achieve marked increases of tendon stiffness.

(Mersmann et al., 2017a. Used with permission by Frontiers Media SA.)

and 100° of knee flexion in a parallel squat; 0° = full extension). In these exercises, the overall movement duration should be increased (e.g. to ~6 s) to provide the 3 s high strain application. A low movement velocity and controlled movement behavior are essential to decrease peak forces in the joints (Bergmann et al., 2004). For exemplary exercises, the reader is referred to the supplementary material of Mersmann et al. (2017a). As the tendon already responds to adequate mechanical loading during childhood (Waugh et al., 2014), and the major turnover of

the tendon occurs until the end of adolescence (Heinemeier et al., 2013), it may be beneficial to start implementing specific tendon exercises in the training routines early in the athletic career. It is now well established that guided strength training even with high loads is not only safe for children and adolescents, but even reduces the risk for musculoskeletal injuries (Faigenbaum et al., 2016; Myers et al., 2017). However, as the tendon remains sensitive to exercise even at old age (McCrum et al., 2018), it never seems too late to start training one's tendons.

References

- Arampatzis, A., Karamanidis, K. & Albracht, K. (2007) Adaptational responses of the human Achilles tendon by modulation of the applied cyclic strain magnitude. *J Exp Biol.* 210: 2743–2753.
- Arampatzis, A., Peper, A., Bierbaum, S. & Albracht, K. (2010) Plasticity of human Achilles tendon mechanical and morphological properties in response to cyclic strain. *J Biomech.* 43: 3073–3079.
- Arnoczky, S.P., Lavagnino, M., Whalton, J.H. & Hoonjan, A. (2002) In situ cell nucleus deformation in tendons under tensile load; a morphological analysis using confocal laser microscopy. *J Orthop Res.* 20: 29–35.
- Arya, S. & Kulig, K. (2010) Tendinopathy alters mechanical and material properties of the Achilles tendon. *J Applied Physiol.* 108: 670–675.
- Bergmann, G., Graichen, F. & Rohlmann, A. (2004) Hip joint contact forces during stumbling. *Langenbeck Arch Surg.* 389: 53–59.
- Bohm, S., Marzilger, R., Mersmann, F., Santuz, A. & Arampatzis, A. (2018) Operating length and velocity of human vastus lateralis muscle during walking and running. *Sci Rep.* 8: 5066.
- Bohm, S., Mersmann, F. & Arampatzis, A. (2015) Human tendon adaptation in response to mechanical loading: a systematic review and meta-analysis of exercise intervention studies on healthy adults. *Sports Med Open.* 1: 7.
- Bohm, S., Mersmann, F., Tettke, M., Kraft, M. & Arampatzis, A. (2014) Human Achilles tendon plasticity in response to cyclic strain: effect of rate and duration. *J Exp Biol.* 217: 4010–4017.

Chapter 5

- Bojsen-Møller, J., Magnusson, S.P., Rasmussen, L.R., Kjaer, M. & Aagaard, P. (2005) Muscle performance during maximal isometric and dynamic contractions is influenced by the stiffness of the tendinous structures. *J Appl Physiol*. 99: 986–994.
- Clemmer, J., Liao, J., Davis, D., Horstemeyer, M.F. & Williams, L.N. (2010) A mechanistic study for strain rate sensitivity of rabbit patellar tendon. *J Biomech*. 43: 2785–2791.
- Couppé, C., Svensson, R.B., Silbernagel, K.G., Langberg, H. & Magnusson, S.P. (2015) Eccentric or concentric exercises for the treatment of tendinopathies? *Journal of Orthopaedic and Sports Physical Therapy*. 1–25.
- Cribb, A.M. & Scott, J.E. (1995) Tendon response to tensile stress: an ultrastructural investigation of collagen:proteoglycan interactions in stressed tendon. *J Anat*. 187: 423–428.
- De Monte, G., Arampatzis, A., Stogianari, C. & Karamanidis, K. (2006) *In vivo* motion transmission in the inactive gastrocnemius medialis muscle-tendon unit during ankle and knee joint rotation. *J Electromyogr Kinesiol*. 16: 413–422.
- Depalle, B., Qin, Z., Shefelbine, S.J. & Buehler, M.J. (2015) Influence of cross-link structure, density and mechanical properties in the mesoscale deformation mechanisms of collagen fibrils. *J Mech Behav Biomed Mater*. 52: 1–13.
- Faigenbaum, A.D., Lloyd, R.S., MacDonald, J. & Myer, G.D. (2016) Citius, altius, fortius: beneficial effects of resistance training for young athletes: narrative review. *Br J Sports Med*. 50: 3–7.
- Fouré, A., Nordez, A. & Cornu, C. (2013) Effects of eccentric training on mechanical properties of the plantar flexor muscle-tendon complex. *J Appl Physiol*. 114: 523–537.
- Freitas, S. R., Mendes, B., Le Sant, G., Andrade, R. J., Nordez, A., & Milanovic, Z. (2018) Can chronic stretching change the muscle-tendon mechanical properties? A review. *Scandinavian Journal of Medicine & Science in Sports*. 28(3): 794–806. <http://doi.org/10.1111/sms.12957>
- Gabbett, T.J. (2016) The training— injury prevention paradox: should athletes be training smarter and harder? *Br J Sports Med*. 50: 273–280.
- Gumucio, J.P., Phan, A.C., Ruehlmann, D.G., Noah, A.C. & Mendias, C.L. (2014) Synergist ablation induces rapid tendon growth through the synthesis of a neotendon matrix. *J Appl Physiol*. 117: 1287–1291.
- Heinemeier, K.M., Schjerling, P., Heinemeier, J., Magnusson, S.P. & Kjaer, M. (2013) Lack of tissue renewal in human adult Achilles tendon is revealed by nuclear bomb (14)C. *FASEB J*. 27: 2074–2079.
- Ingelmark, B.E. (1945) Über den Bau der Sehnen während verschiedener Altersperioden und unter verschiedenen funktionellen Bedingungen. Eine Untersuchung der distalen Sehnen des Musculus biceps brachii und Musculus semitendineus des Menschen sowie der Achillessehnen von Kaninchen und weissen Mäusen. *Uppsala Läkareförenings förhandlingar*. 50: 357–395.
- Joshi, S.D. & Webb, K. (2008) Variation of cyclic strain parameters regulates development of elastic modulus in fibroblast/substrate constructs. *J Orthop Res*. 26: 1105–1113.
- Karamanidis, K., Arampatzis, A. & Mademli, L. (2008) Age-related deficit in dynamic stability control after forward falls is affected by muscle strength and tendon stiffness. *J of Electromyogr Kinesiol*. 18: 980–989.
- Kastelic, J., Palley, I. & Baer, E. (1980) A structural mechanical model for tendon crimping. *J Biomech*. 13: 887–893.
- Kjaer, M. & Heinemeier, K.M. (2014) Eccentric exercise: acute and chronic effects on healthy and diseased tendons. *J Appl Physiol*. 116: 1435–1438.
- Kongsgaard, M., Reitelseder, S., Pedersen, T.G., Holm, L., Aagaard, P., Kjaer, M. & Magnusson, S.P. (2007) Region specific patellar tendon hypertrophy in humans following resistance training. *Acta Physiol*. 191: 111–121.
- Kubo, K., Ikebukuro, T., Maki, A., Yata, H. & Tsunoda, N. (2012) Time course of changes in the human Achilles tendon properties and metabolism during training and detraining *in vivo*. *Eur J Appl Physiol*. 112: 2679–2691.
- Kubo, K., Ikebukuro, T., Yata, H., Tsunoda, N. & Kanehisa, H. (2010) Time course of changes in muscle and tendon properties during strength training and detraining. *J Strength Cond Res*. 24: 322–331.
- Kubo, K., Kanehisa, H., Ito, M. & Fukunaga, T. (2001) Effects of isometric training on the elasticity of human tendon structures *in vivo*. *J Appl Physiol*. 91: 26–32.
- LaCroix, A.S., Duenwald-Kuehl, S.E., Lakes, R.S. & Vanderby, R. (2013)

Mechanical loading and adaptive responses of tendinous tissues

- Relationship between tendon stiffness and failure: a metaanalysis. *J Appl Physiol.* 115: 43–51.
- Lavagnino, M., Arnoczky, S.P., Kepich, E., Caballero, O. & Haut, R.C. (2008) A finite element model predicts the mechanotransduction response of tendon cells to cyclic tensile loading. *Biomech Model Mechan.* 7: 405–416.
- Lavagnino, M., Arnoczky, S.P., Tian, T. & Vaupel, Z. (2003) Effect of amplitude and frequency of cyclic tensile strain on the inhibition of MMP-1 mRNA expression in tendon cells: an *in vitro* study. *Connect Tissue Res.* 44: 181–187.
- Lavagnino, M., Wall, M.E., Little, D., Banes, A.J., Guilak, F. & Arnoczky, S.P. (2015) Tendon mechanobiology: Current knowledge and future research opportunities. *J Orthop Res.* 33: 813–822.
- Lian, O.B., Engebretsen, L. & Bahr, R. (2005) Prevalence of jumper's knee among elite athletes from different sports: a cross-sectional study. *Am J Sports Med.* 33: 561–567.
- Malliaras, P., Barton, C.J., Reeves, N.D. & Langberg, H. (2013) Achilles and patellar tendinopathy loading programmes: a systematic review comparing clinical outcomes and identifying potential mechanisms for effectiveness. *Sports Med.* 43: 267–286.
- McCrum, C., Leow, P., Epro, G., König, M., Meijer, K. & Karamanidis, K. (2018) Alterations in leg extensor muscle-tendon unit biomechanical properties with ageing and mechanical loading. *Front Physiol.* 9: 2743.
- Mersmann, F., Bohm, S. & Arampatzis, A. (2017a) Imbalances in the development of muscle and tendon as risk factor for tendinopathies in youth athletes: a review of current evidence and concepts of prevention. *Front Physiol.* 8: 1–18.
- Mersmann, F., Bohm, S., Schroll, A., Marzilger, R. & Arampatzis, A. (2016) Athletic training affects the uniformity of muscle and tendon adaptation during adolescence. *J Appl Physiol.* 121: 893–899.
- Mersmann, F., Charcharis, G., Bohm, S. & Arampatzis, A. (2017b) Muscle and tendon adaptation in adolescence: elite volleyball athletes compared to untrained boys and girls. *Front Physiol.* 8: 613.
- Myers, A.M., Beam, N.W. & Fakhoury, J.D. (2017) Resistance training for children and adolescents. *Transl Pediatr.* 6: 137–143.
- Nikolaidou, M.E., Marzilger, R., Bohm, S., Mersmann, F. & Arampatzis, A. (2017) Operating length and velocity of human M. vastus lateralis fascicles during vertical jumping. *R Soc Open Sci.* 4
- Ozaki, H., Loenneke, J.P., Buckner, S.L. & Abe, T. (2016) Muscle growth across a variety of exercise modalities and intensities: Contributions of mechanical and metabolic stimuli. *Med Hypotheses.* 88: 22–26.
- Roberts, T.J. (2016) Contribution of elastic tissues to the mechanics and energetics of muscle function during movement. *J Exp Biol.* 219: 266–275.
- Robinson, P.S., Lin, T.W., Jawad, A.F., Iozzo, R.V. & Soslowsky, L.J. (2004) Investigating tendon fascicle structure-function relationships in a transgenic-age mouse model using multiple regression models. *Ann Biomed Eng.* 32: 924–931.
- Wang, J. (2006) Mechanobiology of tendon. *J Biomech.* 39: 1563–1582.
- Waugh, C.M., Alktebi, T., de Sa, A. & Scott, A. (2018) Impact of rest duration on Achilles tendon structure and function following isometric training. *Scand J Med Sci Sports.* 28: 436–445.
- Waugh, C.M., Korff, T., Fath, F. & Blazeovich, A.J. (2014) Effects of resistance training on tendon mechanical properties and rapid force production in prepubertal children. *J Appl Physiol.* 117: 257–266.
- Wiesinger, H.-P., Kösters, A., Müller, E. & Seynnes, O.R. (2015) Effects of increased loading on *in vivo* tendon properties: a systematic review. *Med Sci Sports Exerc.* 47: 1885–1895.
- Wren, T.A.L., Lindsey, D.P., Beaupré, G.S. & Carter, D.R. (2003) Effects of creep and cyclic loading on the mechanical properties and failure of human Achilles tendons. *Ann Biomed Eng.* 31: 710–717.

